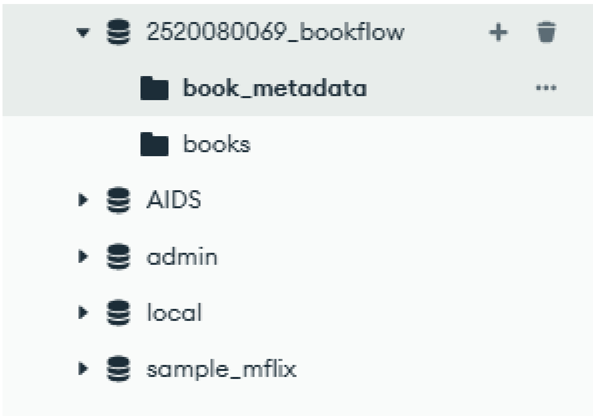
PRACTICAL WEEK3 : MongoDB Atlas

PART 1 : Basics

Query 1 : Create Database and collections

OUTPUT

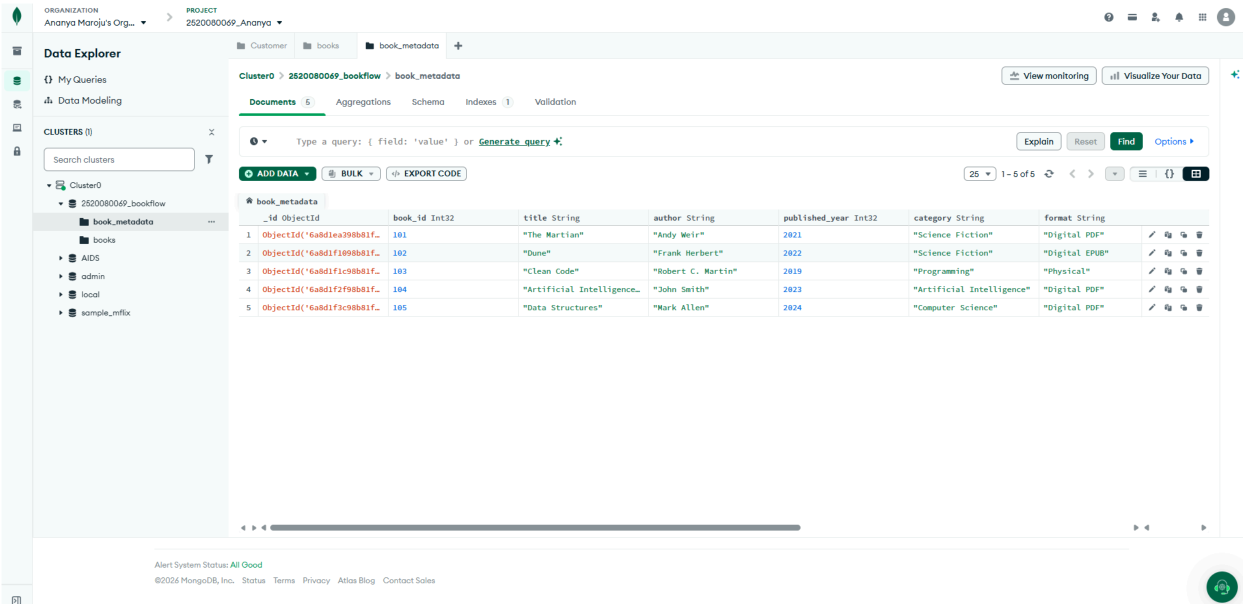


Query 2 : Insert documents

```
{
 "book_id": 101,
 "title": "The Martian",
 "author": "Andy Weir",
 "published_year": 2021,
 "category": "Science Fiction",
 "format": "Digital PDF",
 "file_size": "2.5 MB",
 "member_id": 101,
 "reviews": [
   {
    "member_id": 101,
    "rating": 5,
    "comments": "Excellent science fiction story"
   },
   {
    "member_id": 102,
    "rating": 4,
    "comments": "Very interesting and informative"
   },
   {
    "member_id": 103,
    "rating": 5,
    "comments": "Excellent space adventure"
```

```
    }
  ]
}
```

OUTPUT:

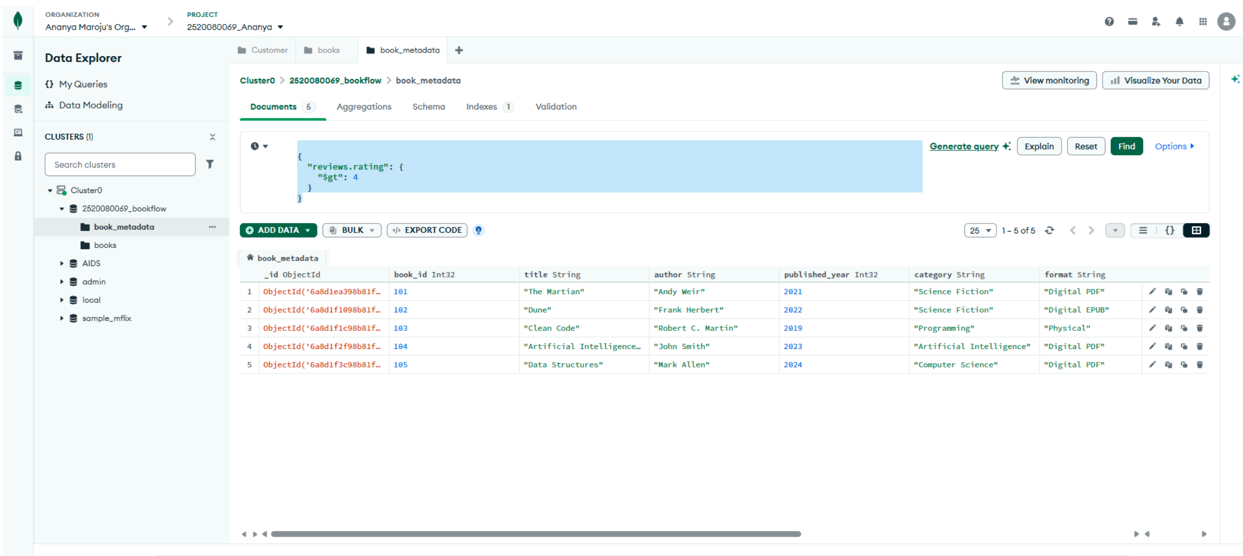


Data Explorer
Cluster0 > 2520080069_bookflow > book_metadata
Documents 5
_id ObjectId | book_id Int32 | title String | author String | published_year Int32 | category String | format String
1 ObjectId('6a8d1ea398b81f… 101 "The Martian" "Andy Weir" 2021 "Science Fiction" "Digital PDF"
2 ObjectId('6a8d1f1098b81f… 102 "Dune" "Frank Herbert" 2022 "Science Fiction" "Digital EPUB"
3 ObjectId('6a8d1f1c98b81f… 103 "Clean Code" "Robert C. Martin" 2019 "Programming" "Physical"
4 ObjectId('6a8d1f2f98b81f… 104 "Artificial Intelligence… "John Smith" 2023 "Artificial Intelligence" "Digital PDF"
5 ObjectId('6a8d1f3c98b81f… 105 "Data Structures" "Mark Allen" 2024 "Computer Science" "Digital PDF"

Query 3 : Comparison Operator

```
{
 "reviews.rating": {
   "$gt": 4
 }
}
```

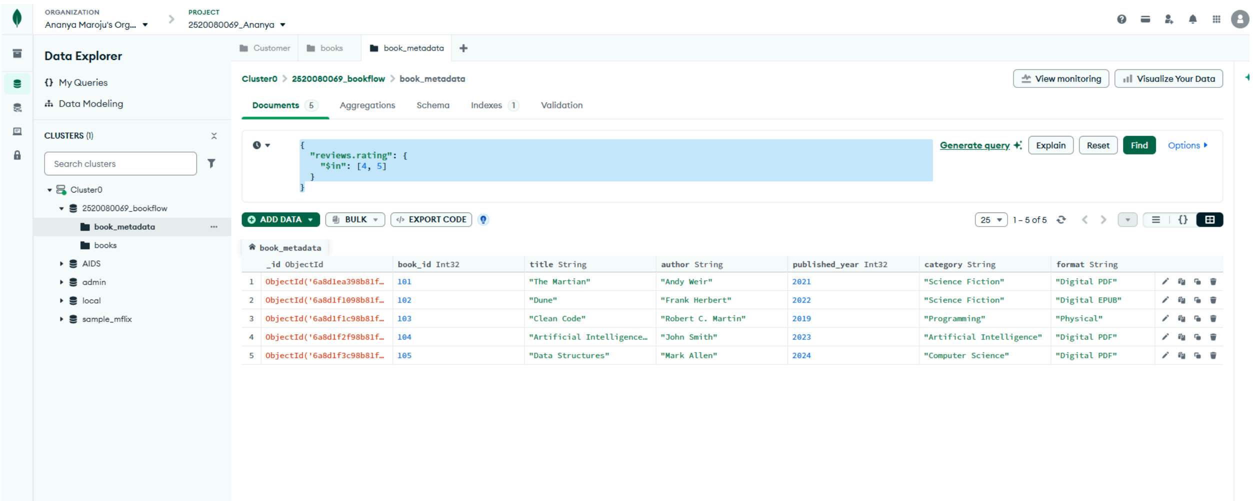
OUTPUT :



Query 4 : Using $IN

```
{
  "reviews.rating": {
    "$in": [4, 5]
  }
}
```

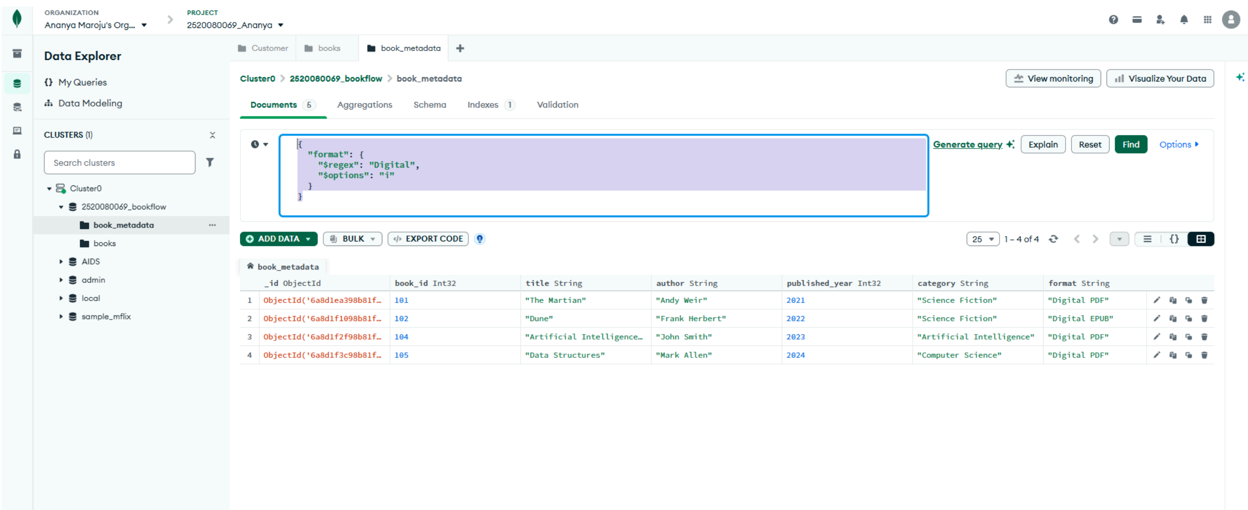
OUTPUT:



Query 5 : Regex query

```
{
  "format": {
    "$regex": "Digital",
    "$options": "i"
  }
}
```

OUTPUT:



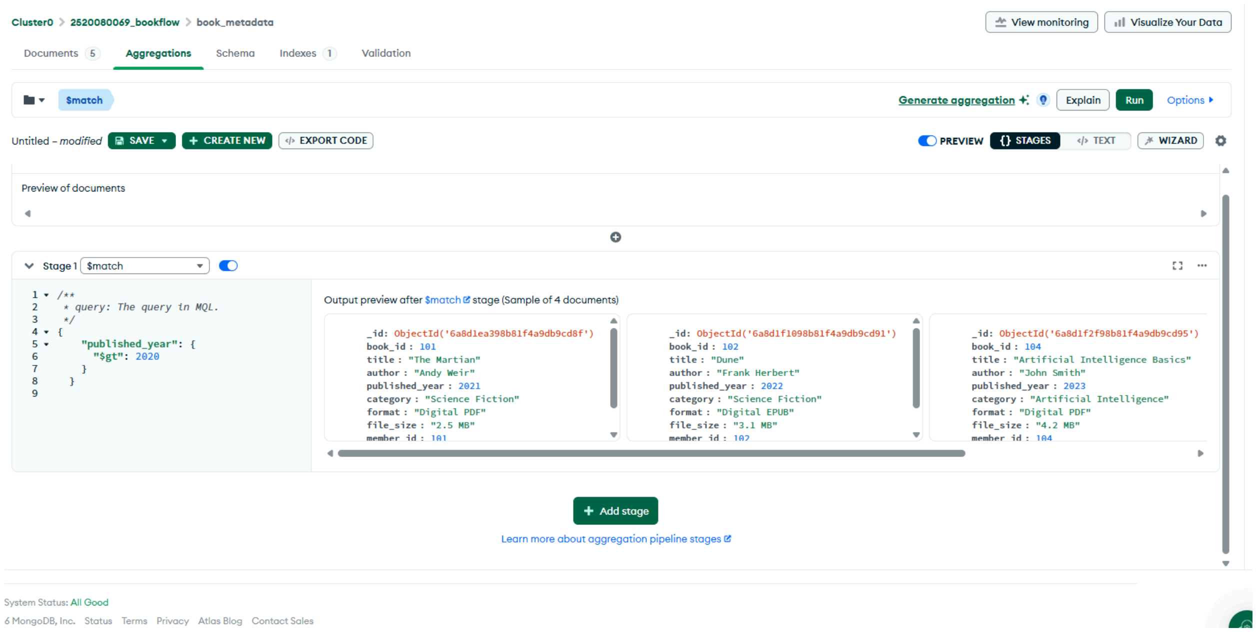
PART 2 : Aggregate pipeline

Query 6 : $MATCH

Stage:

```
{
  "$match": {
    "published_year": {
      "$gt": 2020
    }
  }
}
```

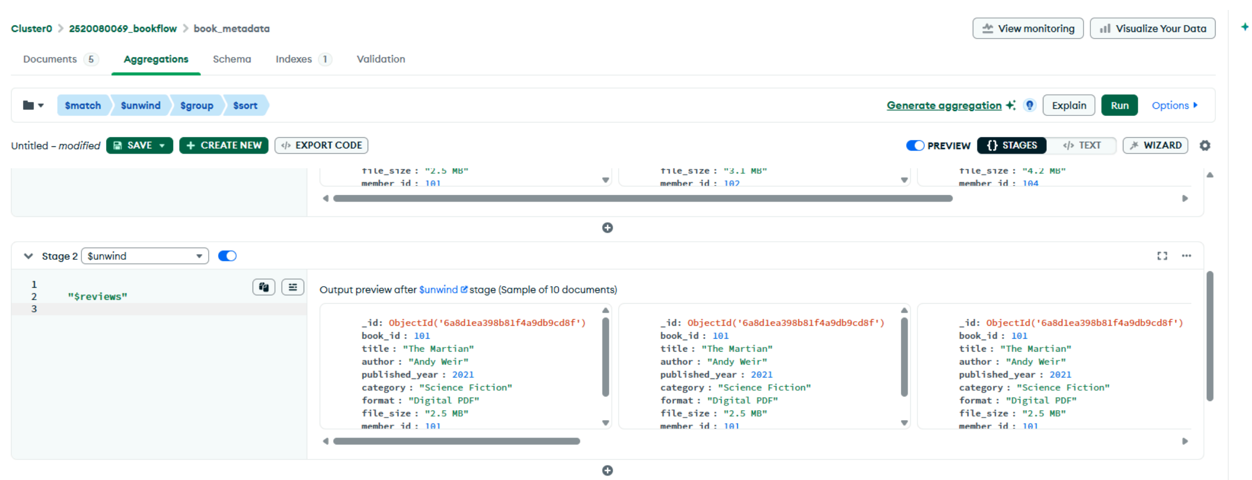
OUTPUT:



Query 7 : $UNWIND

```
{
 "$unwind": "$reviews"
}
```

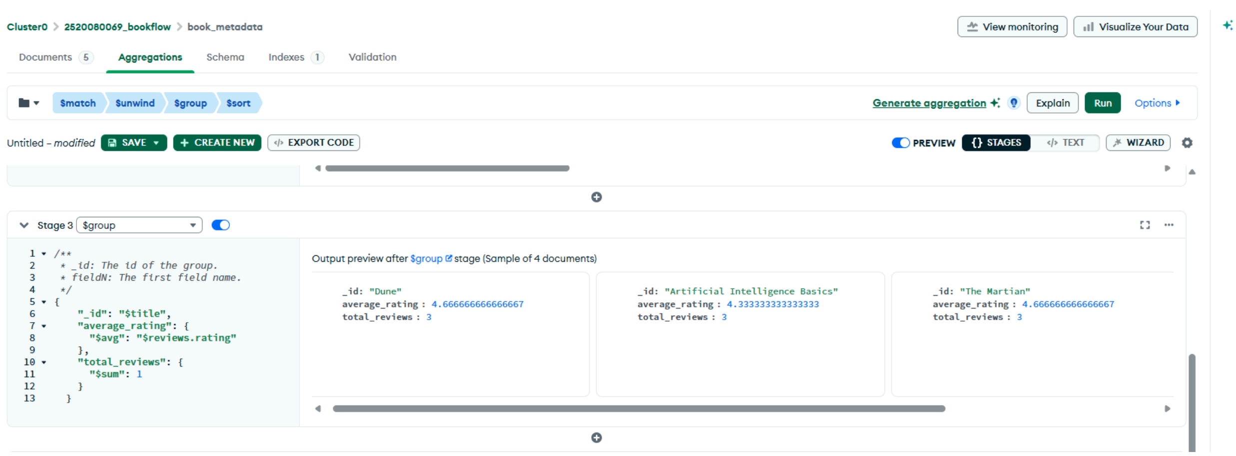
OUTPUT:



Query 8 : $GROUP

```
{
  "$group": {
    "_id": "$title",
    "average_rating": {
      "$avg": "$reviews.rating"
    },
    "total_reviews": {
      "$sum": 1
    }
  }
}
```

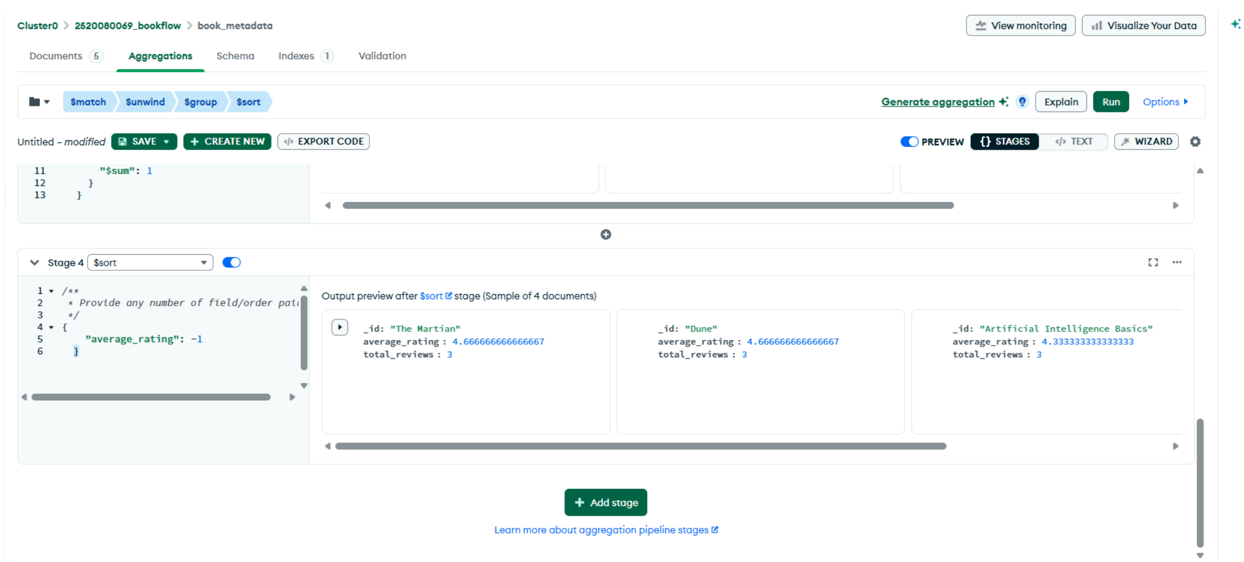
OUTPUT:



Query 9 : $SORT

```
{
  "$sort": {
    "average_rating": -1
  }
}
```

OUTPUT:

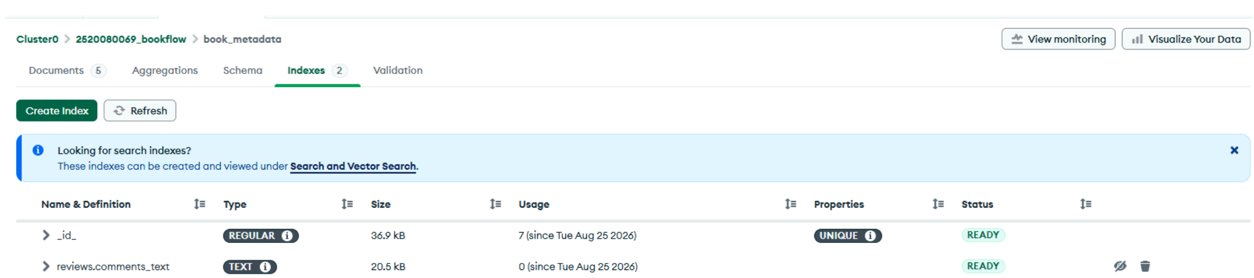


PART 3 : Text Index

Query 10 : Creation of text index

```
db.book_metadata.createIndex(
  {
    "reviews.comments": "text"
  }
)
```

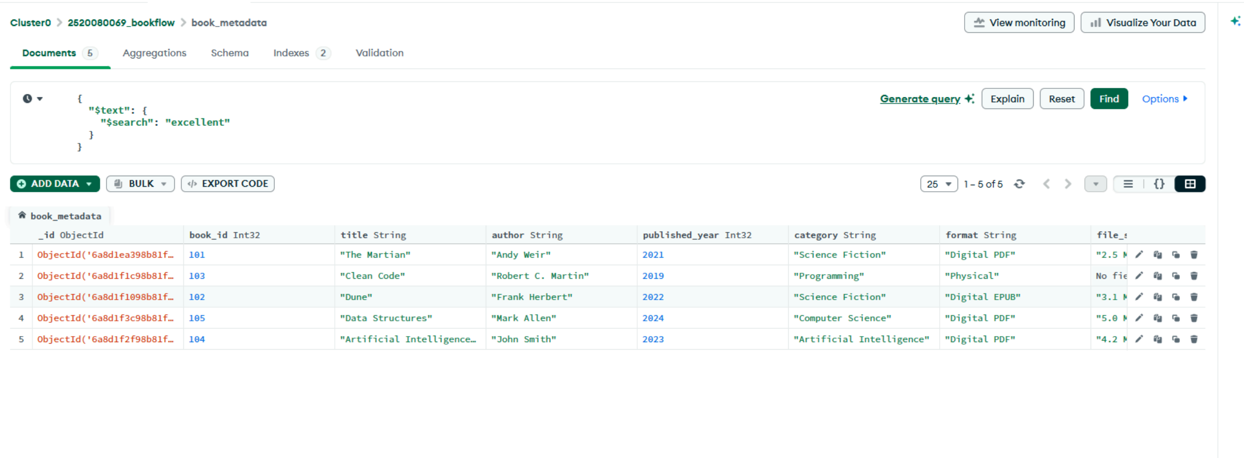
OUTPUT:



Query 11 : Search using text index

```
{
  "$text": {
    "$search": "excellent"
  }
}
```

OUTPUT:

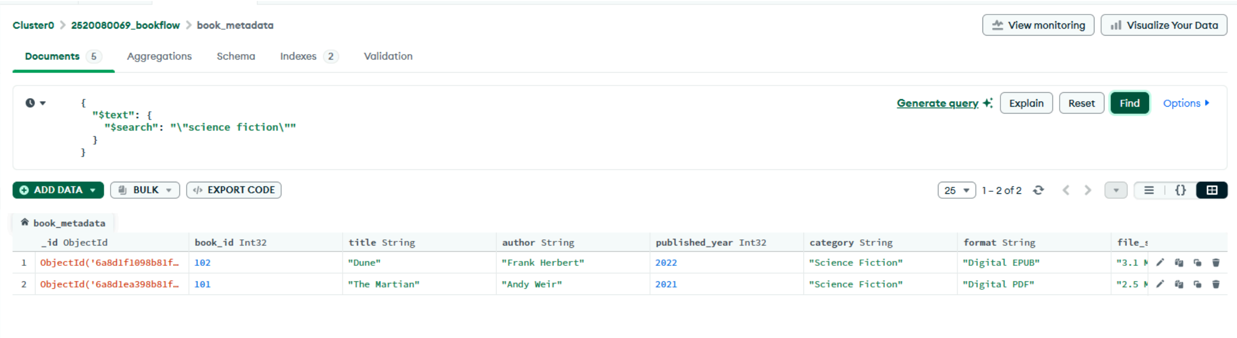


	_id ObjectId	book_id Int32	title String	author String	published_year Int32	category String	format String	file_s
1	ObjectId('6a8d1ea398b81f…	101	"The Martian"	"Andy Weir"	2021	"Science Fiction"	"Digital PDF"	"2.5 M
2	ObjectId('6a8d1f1c98b81f…	103	"Clean Code"	"Robert C. Martin"	2019	"Programming"	"Physical"	No fie
3	ObjectId('6a8d1f1098b81f…	102	"Dune"	"Frank Herbert"	2022	"Science Fiction"	"Digital EPUB"	"3.1 M
4	ObjectId('6a8d1f3c98b81f…	105	"Data Structures"	"Mark Allen"	2024	"Computer Science"	"Digital PDF"	"5.0 M
5	ObjectId('6a8d1f2f98b81f…	104	"Artificial Intelligence…	"John Smith"	2023	"Artificial Intelligence"	"Digital PDF"	"4.2 M

Query 12 : Search for a specific keyword

```
{
  "$text": {
    "$search": "\"science fiction\""
  }
}
```

OUTPUT:



	_id ObjectId	book_id Int32	title String	author String	published_year Int32	category String	format String	file_s
1	ObjectId('6a8d1f1098b81f…	102	"Dune"	"Frank Herbert"	2022	"Science Fiction"	"Digital EPUB"	"3.1 M
2	ObjectId('6a8d1ea398b81f…	101	"The Martian"	"Andy Weir"	2021	"Science Fiction"	"Digital PDF"	"2.5 M

Query 13 : USE explain()

Check whether MongoDB is using an index or performing a collection scan.

```
{
  "$text": {
    "$search": "excellent"
  }
}
```

OUTPUT: